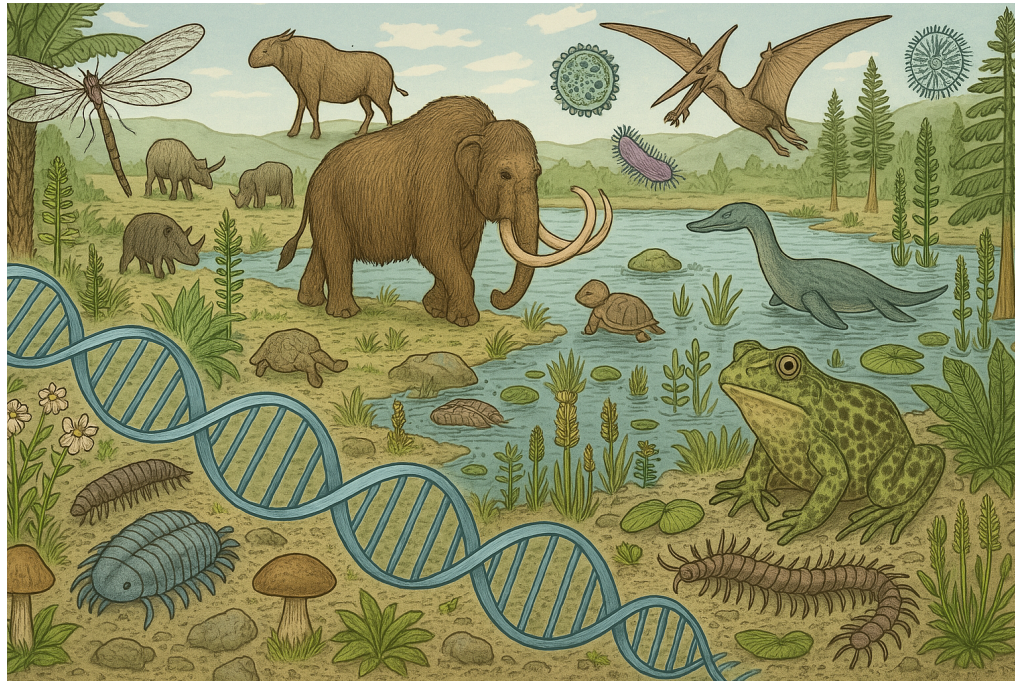


Ancient Metagenomics Analysis

Physalia course, online, 27-29 May 2025

Course outline and practical information

Nikolay Oskolkov, Lund University, NBIS SciLifeLab



About us

Organizer: Carlo Pecoraro, Physalia courses

info@physalia-courses.org

Instructor:

Dr. Nikolay Oskolkov, Lund University, NBIS SciLifeLab

Nikolay.Oskolkov@biol.lu.se



@NikolayOskolkov



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Personal homepage:
<https://nikolay-osolkov.com>

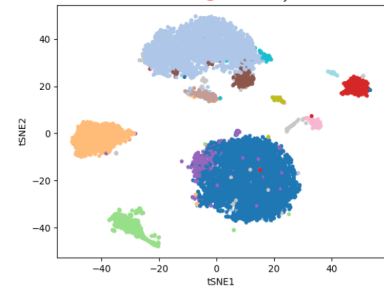
2007 PhD in theoretical physics

2011 medical genomics at Lund University

2016 working at NBIS SciLifeLab, Sweden



Single cell



Biomedical data integration

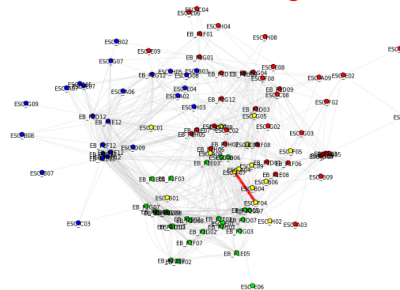
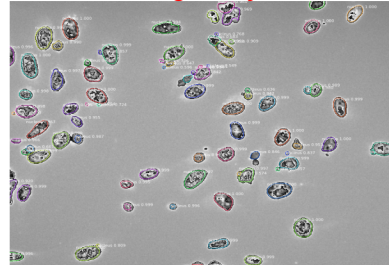
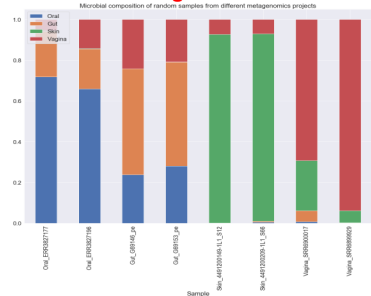


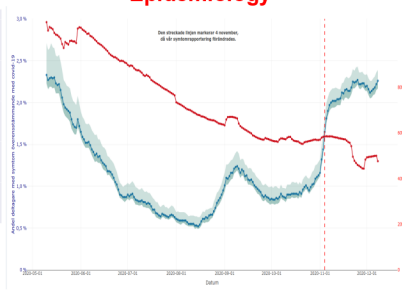
Image analysis



Metagenomics



Epidemiology



Ancient microbiome



METHOD

Open Access

aMeta: an accurate and memory-efficient ancient metagenomic profiling workflow



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[†]Zoé Pochon, Nora Bergfeldt, Anders Götherström, Claudio Mirabello, Per Unneberg, and Nikolay Oskolkov shared authorship.

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Abstract

Analysis of microbial data from archaeological samples is a growing field with great potential for understanding ancient environments, lifestyles, and diseases. However, high error rates have been a challenge in ancient metagenomics, and the availability of computational frameworks that meet the demands of the field is limited. Here, we propose aMeta, an accurate metagenomic profiling workflow for ancient DNA designed to minimize the amount of false discoveries and computer memory requirements. Using simulated data, we benchmark aMeta against a current state-of-the-art workflow and demonstrate its superiority in microbial detection and authentication, as well as substantially lower usage of computer memory.

Keywords: Ancient metagenomics, Pathogen detection, Microbiome profiling, Ancient DNA

Background

Historically, ancient DNA (aDNA) studies have focused on human and faunal evolution and demography, extracting and analyzing predominantly eukaryotic aDNA [1–3]. With the development of next-generation sequencing (NGS) technologies, it was demonstrated that host-associated microbial aDNA from eukaryotic remains, which was previously treated as a sequencing by-product, can provide valuable information about ancient pandemics, lifestyle, and population migrations in the past [4–6]. Modern technologies have made it possible to study not only ancient microbiomes populating eukaryotic hosts, but also sedimentary ancient DNA (sedaDNA), which has rapidly become an independent branch of palaeogenetics, delivering unprecedented information about hominin and animal evolution without the need to analyze historical bones and teeth [7–12]. Previously available in microbial ecology, meta-barcoding methods lack validation and authentication power, and therefore, shotgun metagenomics has become the *de facto* standard in ancient microbiome research [13]. However, accurate detection,

aMeta: an accurate and memory-efficient ancient Metagenomic profiling workflow

snakemake 6.10.0 Tests passing

About

aMeta is a Snakemake workflow for identifying microbial sequences in ancient DNA shotgun metagenomics samples. The workflow performs:

- trimming adapter sequences and removing reads shorter than 30 bp with Cutadapt
- quality control before and after trimming with FastQC and MultiQC
- taxonomic sequence kmer-based classification with KrakenUniq
- sequence alignment with Bowtie2 and screening for common microbial pathogens
- deamination pattern analysis with MapDamage2
- Lowest Common Ancestor (LCA) sequence alignment with Malt
- authentication and validation of identified microbial species with MaltExtract

When using aMeta and / or pre-built databases provided together with the workflow for your research projects, please cite our preprint: <https://www.biorxiv.org/content/10.1101/2022.10.03.510579v1>

Authors

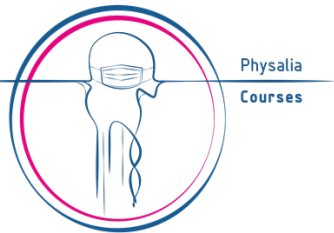
- Nikolay Oskolkov (@LeandroRitter) nikolay.oskolkov@scilifelab.se
- Claudio Mirabello (@clami66) claudio.mirabello@scilifelab.se
- Per Unneberg (@percyfal) per.unneberg@scilifelab.se



<https://github.com/NBISweden/aMeta>

About you

- Name
- University / Institute / Company
- Research interest(s)
- Previous experience with computational analysis and bioinformatics
- Motivation to join the course
- Expectations from the course



Typical analysis methods used in ancient metagenomics

1) Alignment:



2) Classification:



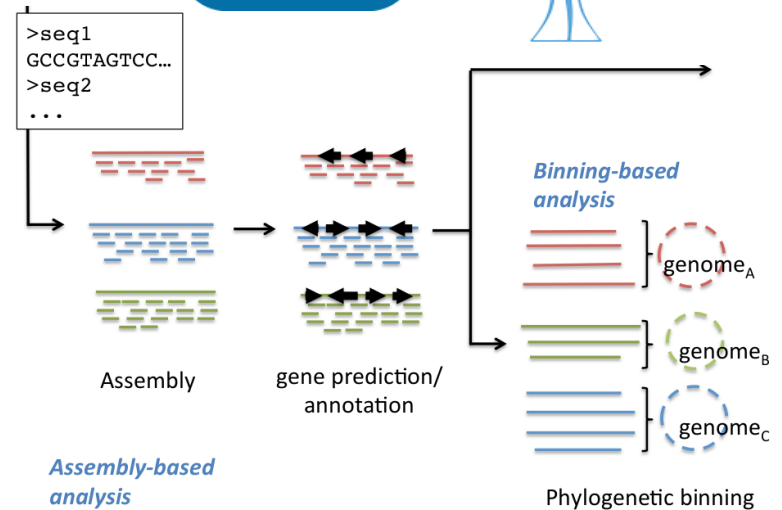
Centrifuge

MetaPhlan

Clark

Reference based:
assume similarity to reference

3) De-novo assembly:



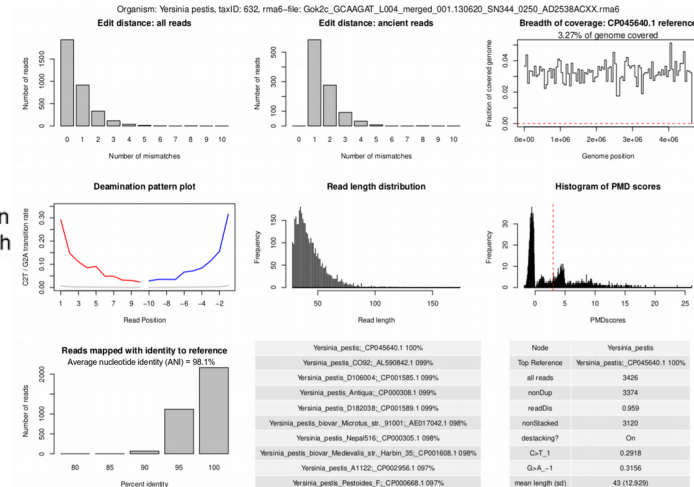
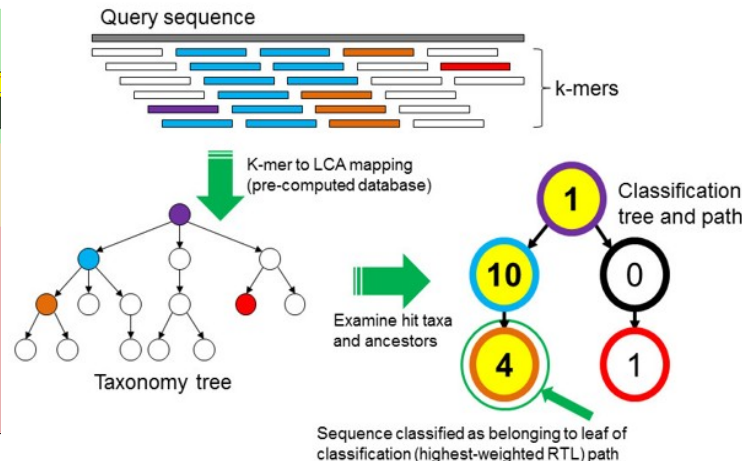
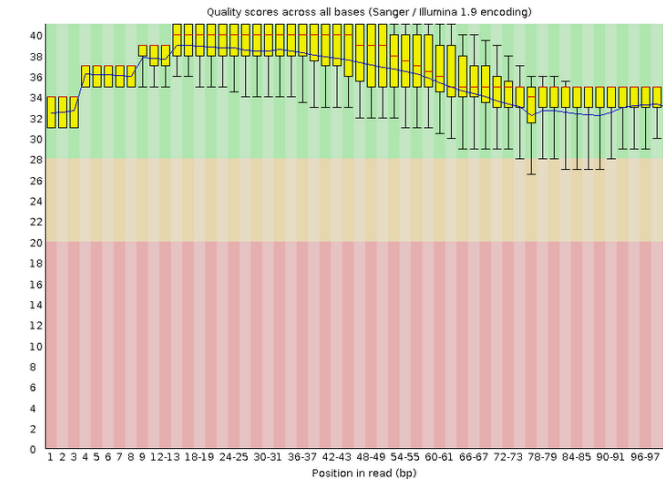
Reference free:
unbiased but challenging

Course outline

Day 1: introduction to ancient microbial and environmental metagenomics, quality control, adapter removal, host removal, taxonomic profiling

Day2: alignment, accessing mapping quality, decontamination, source tracking and authentication analyses, applying LCA to alignments for environmental samples

Day3: aMeta workflow, *de novo* assembly and authentication of assembled contigs



- To ask questions please **raise your hand, unmute yourself and ask**. You can also ask questions in the **zoom chat** or the **slack channel** for this course.
- I would greatly appreciate if you **keep your camera on** as much as possible for better contact and communication.
- The course includes **7 lectures (~1h each) and 7 practicals (~1.5h each)**, there is a **15 min break** after each lecture and each practical
- During practicals, we will use **bash command line tools**. At the end of each practical, I will go through command lines with my explanations.
- The course assumes **some familiarity with basics of bash / R / Python / aDNA**, you do not have to be fluent though. In case any terminology is unclear, please ask, I am ready to provide more detailed explanations.

Practical information: GitHub and Zoom

The course will take place in Zoom from 2 pm to 8 pm (CET, Berlin time)

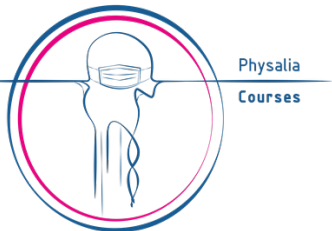
Links to the Zoom room will be posted in the Slack channel

The course GitHub repository containing lectures and exercises is:

https://github.com/NikolayOskolkov/Physalia_AncientMetagenomics_2025

Please bookmark this address! To copy the material to your computer, please do:

git clone https://github.com/NikolayOskolkov/Physalia_AncientMetagenomics_2025



Physalia_AncientMetagenomics_2025

Public

Pin

Unwatch 1

Fork 0

Star 0

master

1 Branch

0 Tags

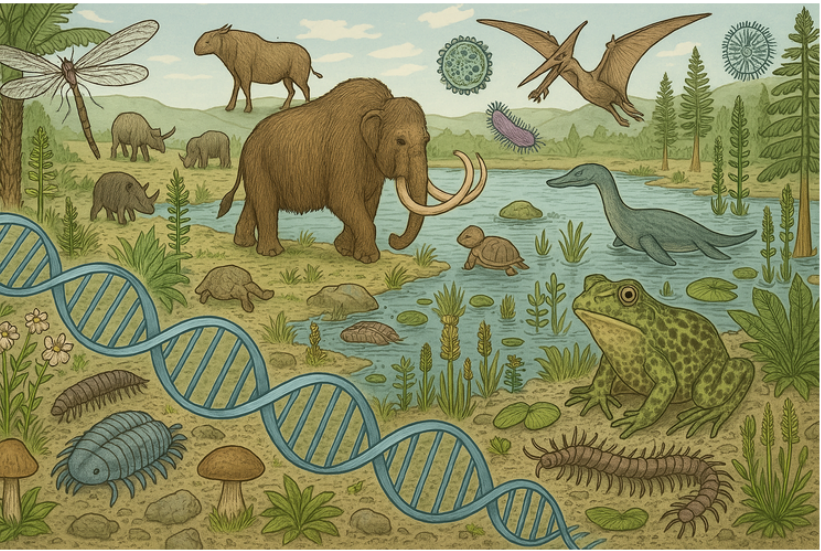
Go to file

Add file

Code

 LeandroRitter	added articles	30370db · 7 hours ago	7 Commits
 articles	added articles		7 hours ago
 practicals	added material		11 hours ago
 slides	added material		11 hours ago
 README.md	added text to README		8 hours ago
 command-line-basics.md	added material		11 hours ago
 logo.png	changed logo		10 hours ago

README



About

No description, website, or topics provided.

Readme

Activity

0 stars

1 watching

0 forks

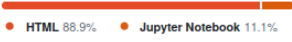
Releases

No releases published
[Create a new release](#)

Packages

No packages published
[Publish your first package](#)

Languages



Suggested workflows

Based on your tech stack

 SLSA Generic generator

Generate SLSA3 provenance for your existing release workflows

Configure

 Jekyll using Docker image

Package a Jekyll site using the jekyll/builder Docker image.

Configure

More workflows

Dismiss suggestions

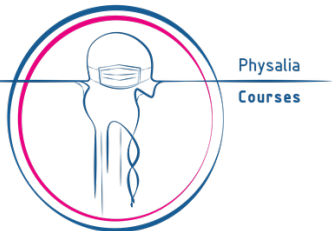
Practical information: Amazon Cloud (AWS EC2)



We will use the Cloud Computing service from Amazon, which we will access via `ssh` (secure shell protocol)

```
ssh -i ameta.pem -X ubuntu@54.244.40.220
```

- The IP address changes every day
- Everyone is given a username, with a `home` and `shared` folders
 - List of usernames can be found in Slack
 - The `shared` folder is copy-only: do not delete, move, rename, or write



To copy remote files and open them locally, please run:

```
scp -i ameta25.pem ubuntu@54.244.40.220:/remote_dir/remote_file /local_dir
```

ActivitiesTerminal

fre 20:31

sv🔊📄

nikolay@dell: ~/Documents/Consultation/Physalia/ancient_metagenomics

FileEditViewSearchTerminalHelp

(base) nikolay@dell:~/Documents/Consultation/Physalia/ancient_metagenomics\$ ssh -i ameta25.pem ubuntu@54.202.33.70

Welcome to Ubuntu 24.04.1 LTS (GNU/Linux 6.8.0-1021-aws x86_64)

* Documentation: <https://help.ubuntu.com>

* Management: <https://landscape.canonical.com>

* Support: <https://ubuntu.com/pro>

System information as of Thu May 22 08:18:12 UTC 2025

System load: 0.0

Processes: 165

Usage of /: 18.8% of 967.64GB

Users logged in: 0

Memory usage: 1%

IPv4 address for enX0: 172.31.36.154

Swap usage: 0%

Expanded Security Maintenance for Applications is not enabled.

63 updates can be applied immediately.

To see these additional updates run: apt list --upgradable

8 additional security updates can be applied with ESM Apps.

Learn more about enabling ESM Apps service at <https://ubuntu.com/esm>

Last login: Thu May 22 08:18:13 2025 from 130.235.244.193

(base) ubuntu@ip-172-31-36-154:~\$ ll

total 396508

drwxr-xr-x	14	ubuntu	ubuntu	4096	May	21	14:59	./
drwxr-xr-x	40	root	root	4096	Jan	8	13:13	../
-rw-r--r--	1	ubuntu	ubuntu	61	Jan	27	11:34	.Xauthority
-rw-r--r--	1	ubuntu	ubuntu	20865	May	21	16:25	.bash_history
-rw-r--r--	1	ubuntu	ubuntu	220	Mar	31	2024	.bash_logout
-rw-r--r--	1	ubuntu	ubuntu	4333	Jan	8	11:11	.bashrc
drwx-----	8	ubuntu	ubuntu	4096	May	21	12:52	.cache/
drwxrwxr-x	2	ubuntu	ubuntu	4096	Jan	9	12:13	.conda/
drwx-----	6	ubuntu	ubuntu	4096	Jan	28	14:35	.config/
drwxrwxr-x	4	ubuntu	ubuntu	4096	May	21	12:52	.java/
-rw-r--r--	1	ubuntu	ubuntu	40	Jan	28	11:48	.lessht
drwxrwxr-x	3	ubuntu	ubuntu	4096	Jan	8	11:24	.local/
drwxrwxr-x	3	ubuntu	ubuntu	4096	May	21	12:51	.parallel/
-rw-r--r--	1	ubuntu	ubuntu	807	Mar	31	2024	.profile
-rw-r--r--	1	ubuntu	ubuntu	11	Jan	8	10:30	.python_history
drwx-----	2	ubuntu	ubuntu	4096	Jan	8	09:52	.ssh/
-rw-r--r--	1	ubuntu	ubuntu	0	Jan	8	10:12	.sudo_as_admin_successful
-rw-r--r--	1	ubuntu	ubuntu	16260	Jan	28	18:58	.viminfo
-rw-rw-r--	1	ubuntu	ubuntu	381	May	21	15:28	.wget-hsts
-rw-rw-r--	1	ubuntu	ubuntu	147784736	Jan	8	11:08	Miniconda3-latest-Linux-x86_64.sh
drwxr-xr-x	3	ubuntu	ubuntu	4096	Jan	8	13:26	R/
drwxr-xr-x	4	ubuntu	ubuntu	4096	May	21	15:04	Share/
drwxrwxr-x	12	ubuntu	ubuntu	4096	May	21	12:44	aMeta/
-rw-rw-r--	1	ubuntu	ubuntu	1318204	Apr	21	2020	libssl1.1_1.1.1f-1ubuntu2_amd64.deb
drwxrwxr-x	19	ubuntu	ubuntu	4096	Jan	8	11:11	miniconda3/
-rw-r--r--	1	ubuntu	ubuntu	574	Jan	13	10:57	r_packages_install.R
-rw-rw-r--	1	ubuntu	ubuntu	128381702	Dec	16	19:50	rstudio-server-2024.12.0-467-amd64.deb
-rw-rw-r--	1	ubuntu	ubuntu	128381702	Dec	16	19:50	rstudio-server-2024.12.0-467-amd64.deb.1
drwxrwxr-x	1	ubuntu	ubuntu	285	Jan	8	14:42	set_passwords.sh*